

objectives

- conduct a case study of Austronesian voice and its relationship to adverbial and serial verbs in Formosan
- propose of mixed A/A' analysis, where the locus of spell-out for voice morphology is the categorizing heads

1 Austronesian voice

The *Austronesian voice system* (aka “Austronesian focus”, “Philippine-type voice”, “symmetrical voice”) is characterized by morphological voice alternations on verbs that designate which argument is the *pivot* of a clause.

(1) properties of Austronesian voice systems:

- voice morphology*: morphological alternations on the verb reflects the choice of pivot
- clausal subject*: the pivot is realized in a particular morphological form and/or structural position
- extraction restriction*: the pivot is the sole argument that can be A'-extracted
- verb initiality*: the verb is sentence initial, and hosts pronominal clitic attachment
- non-pivot arguments*: all non-pivots receive fixed marking depending on their grammatical function

This system is found in the Austronesian languages of Taiwan, the Philippines, Borneo, Sulawesi, and Madagascar; similar systems have also been reported in Dinka (van Urk 2015) and Khanty (Colley & Privoznov 2020).

(2) voice alternations (Paiwan; Ferrell 1979)

- q(**m**)alup [a caucau] tua vavuy i gadu tua vulu actor voice (AV)
 <AV>hunt NOM man OBL pig LOC mountain OBL spear
- qalup-**en** nua caucau [a vavuy] i gadu tua vulu patient voice (PV)
 hunt-PV ERG man NOM pig LOC mountain OBL spear
- qalup-**an** nua caucau tua vavuy [a gadu] tua vulu locative voice (LV)
 hunt-LV ERG man OBL pig NOM mountain OBL spear
- si**-qalup nua caucau tua vavuy i gadu [a vulu] instrumental voice (IV)
 IV-hunt ERG man OBL pig LOC mountain NOM spear
 ‘The man hunts wild pigs in the mountains with a spear.’

(3) A'-extraction limited to pivot argument (Tgdaya Seediq; fieldwork)

- [ima] wada **m**-ekan sari nii? who PST AV-eat taro this
 ‘Who ate this taro?’
- *[ima] wada puq-**un** sari nii? who PST eat-PV taro this
- [maan] puq-**un** na Mahung? what eat-PV ERG Mahung
 ‘What will Mahung eat?’
- *[maan] **m**-ekan (ka) Mahung? what AV-eat NOM Mahung

Austronesian voice systems are distinct from Indo-European voice systems in the following: 1) the core arguments of the verb are obligatory, 2) there are up to four voice distinctions, where non-core arguments can serve as pivot.

There have been various theoretical approaches to Austronesian voice systems:

- argument structure and valence (de Guzman 1978, Aldridge 2004, 2012): AV constructions are intransitive and pivots are absolutive arguments, extraction patterns are due to *ergative extraction constraint* (EEC)
- active vs. passive (Guilfoyle et al. 1992, Kroeger 1993, Wurmbrand and Shimamura 2017): akin to active vs. passive or direct vs. inverse constructions, extraction patterns due to case assignment of each voice form
- information structure (Pearson 2005, Rackowski & Richards 2005, Chen 2017): pivots share properties with topics, e.g. definiteness, binding and reconstruction, competition with wh-extraction and topicalization
- mixed A/A' (Legate 2011, Aldridge 2017, Erlewine 2018, Colley and Privoznov 2020): combines various insights above of A and A' properties, composite probe from joint C and T head that bundles their functions

2 What is an (ad)verb?

Since voice morphology surfaces on verbs, it is helpful to define what counts as a verb. This is complicated by the fact that ordered adverbs can share most if not all of the properties of verbs in Formosan languages.

- (4) properties of verbs in Formosan languages
- inflects for tense, aspect, mood (TAM)
 - hosts voice and derivational morphology
 - can stand alone and select arguments
 - occupies sentence initial position
 - hosts pronominal clitic attachment
- (5) voice and TAM morphology on verbs and adverbs (Kavalan: Chang 2006)
- | | |
|---|---|
| <p>a. t⟨em⟩ayta=ti=iku tu sulal
 ⟨AV⟩see=PFV=1SG.NOM OBL book
 ‘I read a book.’</p> <p>b. etung-an=na=pa ya babuy
 kill-PV=3SG.ERG=FUT NOM pig
 ‘He will kill the pig.’</p> | <p>c. paqanas=pa=iku pasaqay tu qRitun
 slow.AV=FUT=1SG.NOM drive.AV OBL car
 ‘I will drive the car slowly.’</p> <p>d. pataz-an=ku=ti s⟨em⟩upas ya qRitun
 often-PV=1SG.ERG=PFV ⟨AV⟩buff NOM car
 ‘I often buffed the car.’</p> |
|---|---|

Adverbs differ from verbs, however, in their ability to stand alone (for the most part). They occur in constructions termed *adverbial verb constructions* (AVC) which are similar to *serial verb constructions* (SVC) in these languages:

- complementation: not adjunction or conjunction, as evidenced by extraction restrictions and clitic position

- (6) true adjuncts but not adverbial verbs can be extracted (Tgdaya Seediq: fieldwork)
- [klaali] **ka** m-n-ekan=nami ido dhequy ma q⟨n⟩sing-an
 everyday TOP AV-PST-eat=1PL.EXCL rice glutinous and ⟨PST⟩cure-PV
 ‘We ate glutinous rice and cured meat everyday.’
 - *[m-qaras] **ka** m-ekan lama Bakan
 AV-happily TOP AV-eat candy Bakan
 ‘Bakan is happily eating the candy.’
- (7) true conjuncts require two sets of pronominal clitics (Kanakanavu: Wu 2006)
- | | |
|---|---|
| <p>a. m-ukusa=ku (*mata) m-alinsinatu
 AV-go=1SG.NOM and AV-study
 ‘I go study.’</p> | <p>b. m-ukusa=ku *(mata) m-alinsinatu=ku
 AV-go=1SG.NOM and AV-study=1SG.NOM
 ‘I go and I study.’</p> |
|---|---|

- monoclausal structures: single voice and TAM specification, single set of clitics that undergoes clitic climbing to the matrix (ad)verb, phenomena akin to long passive with voice morphology on matrix (ad)verb

- (8) single voice and TAM specification (Tgdaya Seediq: fieldwork)
- wada=mu { bleq-**un** m-ahu / *bleq-**un** phe-**un** / *bleyaq phe-**un** } lukus
 PST=1SG.ERG well-PV AV-wash / well-PV wash-PV / well.AV wash-PV clothes
 ‘I washed the clothes well.’
 - tena m-(**n**-)usa=ku m-(***n**-)angal sama
 already AV-PST-go=1SG.NOM AV-PST-take vegetable
 ‘I already went to forage vegetables.’
- (9) long passive with embedded object promotion (Mayrinax Atayal: Yu 2008, Chen 2010)
- si-punabak=mi**=la cokang ‘i’ k⟨um⟩awah [ku’ kurol]
 IV-carefully=1SG.ERG=ASP very LNK ⟨AV⟩open NOM box
 ‘I’ve already carefully opened the box.’
 - naqaru-**un** ‘i’ t⟨um⟩uting ni yumin [ku’ bawaq]
 finish-PV LNK ⟨AV⟩beat ERG Yumin NOM pig
 ‘Yumin finished killing the pigs.’

How should we analyze the constructions in (9), if Austronesian voice systems are information structural?

3 Case study: Formosan adverbial and serial verbs

A list of adverbial verbs in Formosan languages is given below. The order is consistent with conventional theories of adverb hierarchy (Cinque 1999, Ernst 2001), whether we believe it to be derived syntactically or semantically.

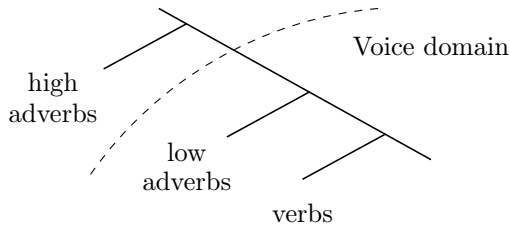
- (10) adverbial verbs in Formosan languages
- a. speaker-oriented adverbs:
 - (i) speech act: *frankly, honestly, allegedly, apparently*
 - (ii) evaluative: *fortunately, unfortunately, pitifully, surprisingly*
 - (iii) epistemic: *perhaps, possibly, surely, probably*
 - b. aspectual adverbs:
 - (i) frequency: *always, sometimes, often, occasionally*
 - (ii) others: *again, first, twice, at last, just, almost*
 - c. event-oriented adverbs:
 - (i) agentive adverbs: *intentionally, diligently, calmly*
 - (ii) manner adverbs: *slowly, carefully, violently, quickly, well, lightly, alone*
- (11) serializing verbs in Formosan languages
- a. auxiliaries: PAST, FUTURE, PROGRESSIVE, IRREALIS, NEGATIVE, HABITUAL
 - b. restructuring verbs: *go, come, start, stop, finish, want, know, try, use, take, stay*

3.1 Verbal properties

However, verbal properties are not uniform across all adverbs. There is a split between high adverbs on the one hand and low adverbs and verbs on the other, corresponding to what lies above vs. below the Voice domain.

- (12) adverbial and lexical verbs, and their verbal properties
- | | | | | |
|------|---|--------------------------------|---|---|
| high | { | speaker-oriented
aspectual | ↑ | PV morphology, sentence initiality,
clitic attachment, (TAM morphology) |
| low | { | event-oriented
lexical verb | ↑ | AV/LV/IV morphology, linkers, stand alone,
prefix concord, (derivational morphology) |

- (13) adverb location with respect to the Voice domain



The distribution of multiple verbal properties coincides with this split, which suggests that it is derived from the structural position of these adverbs (as opposed to the idea that all adverbs are located within the Voice domain).

- overt AV morphology can only be found on low adverbs and lexical verbs (in some languages on verbs only), with the exception of some idiosyncratically null forms, while high adverbs always surface bare in AV form

- (14) (ad)verbs in AV forms (Tgdaya Seediq; fieldwork)
- a. high adverbs: *pgluwe* ‘unfortunately’, *knteetu* ‘often’, *ncugil* ‘sometimes’
 - b. low adverbs: *m-burux* ‘alone’, *g(m)ruus* ‘secretly’, *thuwe* ‘slowly’, *m-qaras* ‘happily’
 - c. lexical verbs: *m-osa* ‘go’, *m-kela* ‘know’, *m-eyah* ‘come’, *prading* ‘start’
- (15) (ad)verbs in PV forms (Tgdaya Seediq; fieldwork)
- a. high adverbs: *pglwe-un* ‘unfortunately’, *kntte-un* ‘often’, *ncgil-un* ‘sometimes’
 - b. low adverbs: *burux-un* ‘alone’, *grus-un* ‘secretly’, *thwe-un* ‘slowly’, *gras-un* ‘happily’
 - c. lexical verbs: *sa-un* ‘go’, *kela-un* ‘know’, *yah-un* ‘come’, *prding-un* ‘start’

- overt LV/IV morphology can only be found on low adverbs and lexical verbs (in some languages on verbs only), e.g. in Tsou auxiliaries and adverbs show up in an underspecified PV form in LV and IV constructions

(16) reduced voice distinction in all adverbs and auxiliaries (Tsou: Chang 2009)

- a. **mo-si asonu mo-si** ta pangka to emi 'o amo actor voice (AV)
AV-3SG possibly.AV AV-put OBL table OBL wine NOM father
'Father possibly put the wine on the table.'
- b. **i-si ason-a si-a** ta pangka to amo 'o emi patient voice (PV)
NAV-3SG possibly-PV put-PV OBL table ERG father NOM wine
'Father possibly put the wine on the table.'
- c. **i-si ason-a si-i** ta amo ta emi 'o pangka locative voice (LV)
NAV-3SG possibly-PV put-LV ERG father OBL wine NOM table
'Father possibly put the wine on the table.'
- d. **i-si ason-a si-eni** ta emi ta amo instrumental voice (IV)
NAV-3SG possibly-PV put-IV OBL wine ERG father
'Father possibly put the wine (for him).'

- *prefix concord* (determined by the lexical verb of the clause) can only be found on low adverbs and lexical verbs, high adverbs are either bare or have adverbial prefixes that are optional on low adverbs

(17) prefix concord *ku-* 'eat' on low but not high adverbs (Saaroa: Li 2009)

- a. **ku-a-muamuare=aku** um-u papa-alemehele
PC{eat}-a-slowly.AV=1SG.NOM AV-eat meat-wild.boar
'I ate wild boar meat slowly.'
- b. (***ku-**)kaarekehle=aku um-a-u papa-alemehele
PC{eat}-often.AV=1SG.NOM AV-a-eat meat-wild.boar
'I often eat wild boar meat.'

- intervening *linkers* can only be found with low adverbs and lexical verbs, or if high adverbs have intervening linkers they are a different form; and this correlates with availability of voice and TAM morphology

(18) linker '*i*' allowed for low but not high adverbs (Mayrinax Atayal: Yu 2008)

- a. m-a'usa' **'i'** ma-panga' cku' 'ulaqi' 'i' yaya'
AV-go LNK AV-carry ACC child NOM mother
'Mother is going to carry the child on her back.'
- b. m-ubagaq **'i'** k(um)alu 'i' Watan
AV-polite LNK (AV)talk NOM Watan
'Watan talked politely.'
- c. kalialia'h (***'i'**) t(um)uting cu' 'ulaqi' 'i' Yumin
often LNK (AV)beat ACC child NOM Yumin
'Yumin beats the child everyday.'
- d. ki'i' (***'i'**) c(in)bu na' ku' bakarowap
probably LNK (PV)shoot INST NOM hunter
'It was probably killed by the hunter.'

There is a question of whether the difference between low and high adverbs is inherent to the adverbs themselves or a consequence of their relative location in terms of other functional elements within the clause.

NOTE: this account should be compatible with variation across Formosan languages; it delineates which properties are possible within a certain domain but does not necessitate that elements within that domain must bear them all.

3.2 Voice patterns

We find two main kinds of voice patterns in Formosan languages, *default voice* and *voice concord* (Wurmbrand & Shimamura 2017, Chang 2017, Wu et al. 2023), which is consistent within languages across AVCs and SVCs.

(19) two voice patterns in Formosan AVCs and SVCs

- a. default voice: first verb has "true" voice morphology, the rest are in "default" AV form
- b. voice concord: all verbs match for consistent forms of voice morphology

- default voice is where the voice morphology on the first (ad)verb indicates which argument will be the pivot, while the remaining (ad)verbs are in AV form, which is the absence of voice features (Chen 2010, Levin 2015)

(20) default voice in AVCs (Tgdaya Seediq: fieldwork)

- a. ciga **m**-n-qaras=ta **m**-imah sino
yesterday AV-PST-happily=1PL.INCL AV-drink alcohol
'We happily drank alcohol yesterday.'
- b. wada=ta gras-**un** **m**-imah [sino kiya] di
PST=1PL.INCL happily-PV AV-drink alcohol that PFV
'We happily drank the alcohol.'

- voice concord is where all (ad)verbs match in voice specification, which does include some surface differences, e.g. prefix vs. infix for AV morphology or underspecified PV morphology on adverbs in LV and IV constructions

(21) voice concord in AVCs (Saisiyat: Yeh 2016)

- a. yako t<**om**>owa'is '(<**om**>)itol, siya 'isa:a' min'itol ila
1SG.NOM <AV>constantly <AV>call 3SG.NOM then wake.up.AV PFV
'I constantly called him, and then he got up.'
- b. nisiya koSa'-en towa'is-**en** singozaw-**en** ni towanay
3SG.ERG say-PV constantly-PV ask-PV ERG sister.in.law
'He did not say it until his sister-in-law constantly asked about.'

Voice morphology (as opposed to TAM morphology or clitic attachment) stands out because of its presence across multiple verbs in the same construction, whether in default voice or voice concord languages.

In addition, in default voice languages there is a mismatch where the initial verb that bears true voice morphology is not necessarily responsible for introducing the pivot; this is particularly interesting for high adverbs.

4 A composite probe account

The central idea is that the categorizing heads of adverbs and verbs are the locus of spell-out for voice morphology, while also being targets of agreement for a *composite probe*. These are the assumptions made:

- categorizing heads: high adverbs have the categorizer *a*, low adverbs and lexical verbs have the categorizer *v*; all share a common *predicative feature* (PRED) that is the mechanism behind adverbs realized as verbs
- clause structure: a Voice projection is present with lexical verbs that introduce external arguments, adverbs (and unaccusative verbs) do not have Voice projections and are in a complementative structure

(22) clause structure in AVCs and SVCs

- a. high adverbs: [CP C ... [_{aP} *a* √adverb [_{VoiceP} Voice [_{vP} *v* √verb ...]]]]
- b. low adverbs: [CP C ... [_{VoiceP} Voice [_{vP} *v* √adverb [_{vP} *v* √verb ...]]]]
- c. restructuring: [CP C ... [_{VoiceP} Voice [_{vP} *v* √verb [_{VoiceP} Voice [_{vP} *v* √verb ...]]]]]

- composite probe: there is a composite probe in C with the features [$\phi \cap \text{TOP}$], that agrees with its closest goal and promotes that argument to pivot status (Erlewine 2018), the agreement features are as below:

(23) agreement features of the composite probe

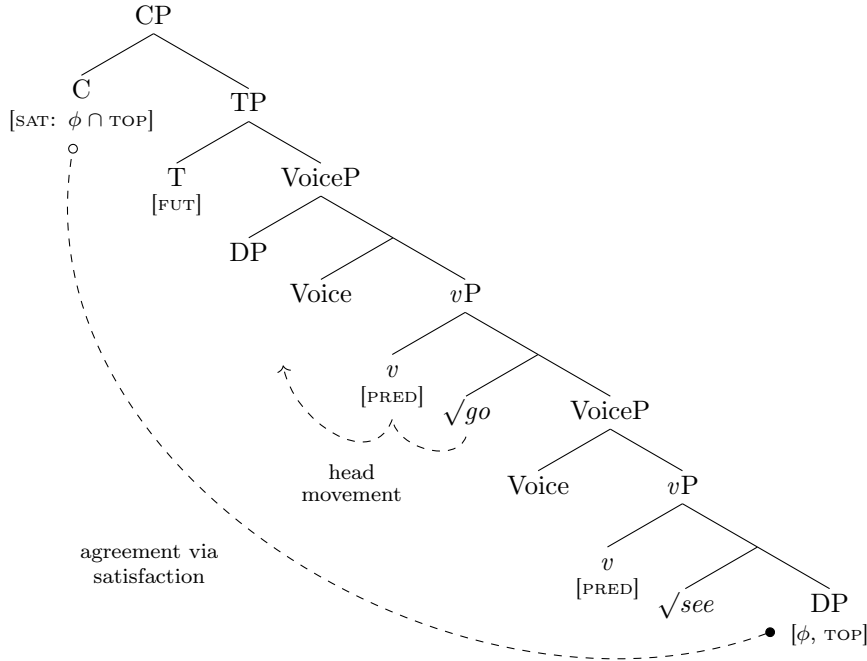
- a. default voice: [SAT: $\phi \cap \text{TOP}$], with head movement
- b. voice concord: [INT: PRED; SAT: $\phi \cap \text{TOP}$]

4.1 Derivation

The difference between default voice and voice concord languages is how voice morphology ends up on highest verb; for the former it is head movement, for the latter it is the extra *interaction condition* [INT: PRED] of the probe.

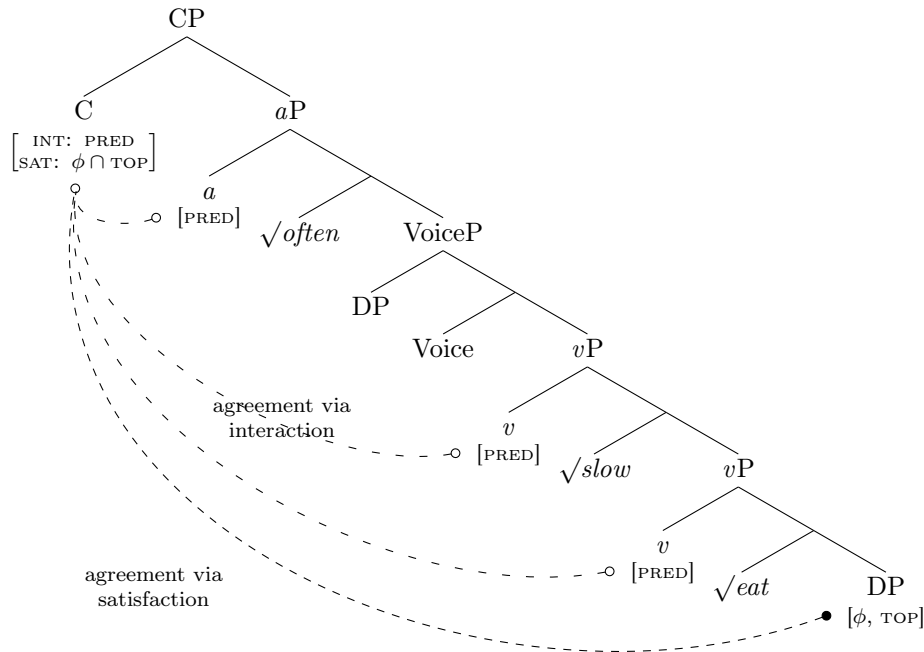
- In default voice languages, the composite probe agrees with the pivot and assigns it nominative case (it also moves to Spec,CP in some but not all languages); the highest (ad)verb undergoes head movement to C

(24) PV construction in default voice language



- In voice concord languages, the composite probe interacts with all elements bearing the PRED feature until it finds the pivot; this interaction agreement is spelled-out as voice morphology on the (ad)verbs

(25) PV construction in voice concord language



If a patient from a higher verb is extracted in a voice concord language, we predict that the probe will be satisfied before reaching the lower verbs, which will not be interacted with and will remain in their default forms:

(26) embedded verb is AV when matrix object is extracted (Tsou: Chang 2010, Lin 2015)

- a. **os**=’o tith-**a** **ma**-hafo to f’ue [’o yungku]
 NAV=1SG use-PV AV-carry OBL sweet.potato NOM basket
 ‘I used the basket to carry sweet potatoes.’

- b. **os**=’o yon-**i** tufku to yusu [’o coca]
 NAV=1SG stay-LV wash.AV OBL clothes NOM yard
 ‘I stayed in the yard washing clothes.’

4.2 Voice features

To address the features that underlie the spell-out of AV and PV morphology, we must first look at case assignment. I assume that Formosan has NOM-ACC alignment, and that “pivot” case is actually a topic marker (Chen 2017).

- The evidence comes from: 1) unaccusative subject has “ergative” case when it is not the pivot, 2) unaccusative verbs have AV morphology when the subject is pivot, 3) pivot and topicalization particles are often the same

(27) the subject of an unaccusative verb has “ergative” case (Seediq: Chen 2017)

- a. h-huqil-an **na**/* $\emptyset$ riso nii ka Paran
 RED-die-LV ERG/OBL young.man this NOM Paran
 ‘This young man will die in Paran.’
- b. s-k⟨n⟩-narux **na**/* $\emptyset$ Temi ka knrudan=na
 IV-⟨PST⟩STAT-sick ERG/OBL Temi NOM age=3SG.GEN
 ‘Temi got sick because of her age.’

Therefore, intransitive subjects and transitive agents share nominative (i.e. “ergative”) case just like in English, and this case gets overridden when the argument becomes the pivot and is marked with topic (i.e. “nominative”) case.

The main claim is that non-AV arises when a non-nominative argument becomes the pivot, which can be derived via: 1) case sensitivity to the pivot (Rackowski & Richards 2005), or 2) bundling of C and T when the subject is pivot.

- AV is the default form of an (ad)verb, PV is the spell-out of A’-agreement features on an (ad)verb
- LV and IV are the spell-out of A’-agreement features with additional locative and instrumental features, which could be either applicative features (Aldridge 2004) or specific case or structural configurations (Chen 2017)

(28) the spell-out of voice morphology

- a. AV: no agreement with composite probe
- b. PV: agreement with composite probe
- (i) LV: and locative features
- (ii) IV: and instrumental features

The vocabulary items are given below. The categorizers *v* and *a* have different default spell-outs, which affects the presence of overt AV morphology, but once they have A’ features they are spelled-out uniformly as PV morphology.

(29) vocabulary items for AV and PV forms in Tgdaya Seediq

- a. “actor voice”: $v \Leftrightarrow m-$ or $\langle m \rangle$
 $a \Leftrightarrow \emptyset$
- b. “patient voice”: $A' \Leftrightarrow -un$

One issue that arises is why categorizing and extraction morphology are in complementary distribution, when the two in theory can coexist; this is a historical consequence of the distinct origins of AV and PV (Begus et al. 2023).

4.3 AV vs. PV

The above analysis of AV and PV explains the distinctions in behavior between the two:

(30) the distinctness of AV and PV morphology

- a. AV is limited to the Voice domain
- b. only AV is sensitive to transitivity and agentivity
- c. only AV is sensitive to the phonological form of the verb root
- d. there are idiosyncratically null forms of AV
- e. the default voice form is AV
- f. AV and PV have distinct historical origins

The idea is that AV morphology reflects features that originate close to the verb root, while PV morphology reflects features further removed; this has consequences on the allomorphic behavior of these forms with respect to the root.

- AV exhibits allomorphy conditioned by the agentivity and phonological form of its root, e.g. in Tgdaya Seediq prefix *m-* becomes infix ⟨*m*⟩ when attached to verbs that are consonant initial and have external arguments
- only AV has idiosyncratically null forms (not to mentioned systematically null forms on high adverbs)

(31) intransitive verbs in AV forms (Tgdaya Seediq; fieldwork)

- unaccusative: *m-huqil* ‘die’, *m-takur* ‘fall’, *m-eeniq* ‘exist’, *dehuk* ‘arrive’
- unergative: *t⟨m⟩alang* ‘run’, *r⟨m⟩engo* ‘talk’, *m-uuyas* ‘sing’, *m-osa* ‘go’, *skiya* ‘fly’
- weather: *q⟨m⟩uyux* ‘rain’, *m-thuda* ‘snow’, *m-bruwa* ‘thunder’, *thido* ‘be sunny’

(32) transitive verbs in AV forms (Tgdaya Seediq; fieldwork)

- vowel initial: *m-ekan* ‘eat’, *m-angal* ‘take’, *m-imah* ‘drink’, *m-etur* ‘kick’
- consonant initial: *s⟨m⟩ipaq* ‘kill’, *c⟨m⟩ebu* ‘shoot’, *k⟨m⟩eruc* ‘cut’, *q⟨m⟩ita* ‘see’

The default form across Formosan languages is always AV, and not another voice form or an entirely different form. This indicates that AV morphology is not only the absence of voice features but has nothing to do with voice at all.

conclusion

- conducted a case study of Austronesian voice and its relationship to adverbial and serial verbs in Formosan
- proposed of mixed A/A’ analysis, where the locus of spell-out for voice morphology is the categorizing heads

future directions:

- more extensive data with stacking of multiple adverbs and verbs, e.g. verbs scoping over adverbs
- a more in depth study of voice across Formosan, and interactions with TAM and derivational morphology
- extend this account to other Austronesian languages, including ones that have active/passive systems too
- a closer look at prefix concord and intervening linkers (and other elements)
- an account of word order (including TAM auxiliaries and negation) given the proposed analysis of voice
- a more detailed analysis which addresses issues of phases and case assignment
- expansion on the different possible kinds of information structure, e.g. topic, focus, nominalization, etc.